

# **WELL - SCOPE: AI-POWERED MENTAL HEALTH MONITOR USING RoBERTa**

A Minor Project Report

Submitted in partial fulfillment of requirement of the Degree  
of

**BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE &  
ENGINEERING**

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## **Report Approval**

The project work "Well-Scope: AI-Powered Mental Health Monitor Using RoBERTa" is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn therein; but approve the "Project Report" only for the purpose for which it has been submitted.

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## Declaration

We hereby declare that the project entitled "Well-Scope: AI-Powered Mental Health Monitor Using RoBERTa" submitted in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science & Engineering, completed under the supervision of Prof. Kamini Vishwakarma and Prof. Priyanka Jain, Assistant Professors, Department of Computer Science & Engineering, Faculty of Engineering, Medicaps University, Indore is an authentic work.

Further, we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

### Signature and name of the students with date:

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## **Certificate**

We, Prof. Kamini Vishwakarma and Prof. Priyanka Jain, certify that the project entitled "Well-Scope: AI-Powered Mental Health Monitor Using RoBERTa" submitted in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science & Engineering by Atharv Untwale, Atharv Namdev, and Atharva Sharma is the record carried out by them under our guidance and that the work has not formed the basis of award of any other degree elsewhere.

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## Abstract

Mental health disorders have become one of the most pressing public health challenges globally, yet remain largely underdiagnosed due to the stigma associated with seeking help and the limited availability of mental health professionals. Social media platforms and personal journaling have emerged as spaces where individuals openly express their emotional states, presenting a significant opportunity for early detection of mental health concerns through natural language processing.

This project presents Well-Scope, an AI-powered mental health monitoring web application that analyzes journal entries and social media text to detect and classify emotional states including depression, anxiety, stress, and positive or neutral moods. Well-Scope employs the RoBERTa-based emotion classification model (j-hartmann/emotion-english-distilroberta-base) fine-tuned on English emotional text, served through a Python FastAPI microservice. The system computes a risk score for each entry and generates actionable wellness insights.

The frontend is built using React.js with a premium Warm Parchment UI theme incorporating Cormorant Garamond and Syne typography. The backend is a Node.js/Express REST API authenticated via Supabase JWT tokens, with user data stored in a PostgreSQL database managed by Supabase. The application additionally features a personalized AI wellness chatbot powered by Llama 3.1 (via Groq API) that remembers user profiles and references the latest journal entries to provide context-aware emotional support. Additional features include a mood calendar heatmap, streak tracker, wellness tips, mood distribution charts, and PDF export.

The system is deployed with the frontend on Vercel and the backend and AI service on Render, demonstrating a fully cloud-native, production-ready mental health monitoring solution.

**Keywords:** Mental Health, RoBERTa, Emotion Detection, Natural Language Processing, MERN Stack, Supabase, Groq, FastAPI, React.js, Journaling, AI Chatbot, Mood Tracking

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## Abbreviations

| Abbreviation | Full Form                                               |
|--------------|---------------------------------------------------------|
| AI           | Artificial Intelligence                                 |
| NLP          | Natural Language Processing                             |
| RoBERTa      | Robustly Optimized BERT Pretraining Approach            |
| BERT         | Bidirectional Encoder Representations from Transformers |
| API          | Application Programming Interface                       |
| REST         | Representational State Transfer                         |
| MERN         | MongoDB, Express, React, Node.js                        |
| JWT          | JSON Web Token                                          |
| CORS         | Cross-Origin Resource Sharing                           |
| RLS          | Row Level Security                                      |
| UI           | User Interface                                          |
| UX           | User Experience                                         |
| DFD          | Data Flow Diagram                                       |
| ER           | Entity Relationship                                     |
| CRUD         | Create, Read, Update, Delete                            |
| PDF          | Portable Document Format                                |
| CDN          | Content Delivery Network                                |
| GPU          | Graphics Processing Unit                                |
| LLM          | Large Language Model                                    |
| SRS          | Software Requirements Specification                     |

# CHAPTER 1: INTRODUCTION

## 1.1 Introduction

Mental health disorders affect more than 970 million people worldwide, according to the World Health Organization (2022). Depression, anxiety, and stress are among the most common conditions, yet a significant proportion of individuals never receive professional diagnosis or treatment. The reasons are multifaceted: social stigma, limited access to mental health professionals, high treatment costs, and, fundamentally, a lack of self-awareness about one's own emotional state over time.

The rapid growth of digital communication has created an unprecedented corpus of emotional expression. People write diary entries, post on social media, and communicate through messaging platforms in ways that often reflect their true psychological states more candidly than clinical interviews. This presents a transformative opportunity: to leverage advances in Natural Language Processing (NLP) and deep learning to detect early signals of mental health distress from text.

Well-Scope is a full-stack web application that addresses this challenge by providing users with a private, secure journaling environment where artificial intelligence analyzes their text entries for emotional content, classifies them into mental health categories, computes a risk score, and tracks mood trends over time. The AI component uses a fine-tuned RoBERTa model for emotion classification. An integrated AI wellness chatbot, powered by the Llama 3.1 language model via the Groq API, provides personalized emotional support informed by the user's journal history and onboarding profile.

## 1.2 Literature Review

Research at the intersection of NLP and mental health has grown substantially in the past decade. Early work by Coppersmith et al. (2014) demonstrated that Twitter data could be used to distinguish users with depression, PTSD, and bipolar disorder from control groups using linguistic features. Their work established the feasibility of social media-based mental health screening.

The introduction of transformer-based models, specifically BERT (Devlin et al., 2018) and its improved variant RoBERTa (Liu et al., 2019), marked a significant advancement in text understanding. These models, pre-trained on large corpora using masked language modeling, demonstrated state-of-the-art performance on a wide range of NLP benchmarks including sentiment analysis and emotion classification tasks.

Hartmann et al. (2022) released the emotion-english-distilroberta-base model on Hugging Face, a distilled version of RoBERTa fine-tuned on multiple emotion datasets. This model classifies text into seven emotion categories (sadness, joy, anger, fear, disgust, surprise, neutral) with high accuracy, making it highly suitable for mental health monitoring applications.

Ji et al. (2021) conducted a comprehensive survey on mental health detection using NLP and machine learning, highlighting that transformer-based models consistently outperform traditional machine learning approaches (SVM, Naive Bayes, Logistic Regression) on mental health classification tasks, particularly when fine-tuned on domain-specific data.

Recent work by Garg et al. (2023) on the CLPsych shared task demonstrated that large language models with appropriate prompting can serve as effective mental health screeners, opening the door to chatbot-based interventions as a complement to traditional therapy. This validates the integration of a conversational AI component in Well-Scope.

### **1.3 Objectives**

The primary objectives of Well-Scope are:

- To develop a secure, user-authenticated web application where individuals can privately submit journal entries for mental health analysis.
- To integrate a pre-trained RoBERTa emotion classification model to detect emotional states (sadness, fear, anger, joy, disgust, surprise, neutral) from free-form text.
- To map detected emotions to mental health categories (depression, anxiety, stress, positive, neutral) and compute a continuous risk score.
- To provide longitudinal mood tracking through interactive charts, a mood calendar heatmap, and a streak tracker to encourage consistent journaling habits.
- To offer personalized wellness tips and an AI chatbot powered by Llama 3.1 (via Groq) that retains user profiles and references journal context for empathetic, context-aware responses.
- To enable users to export their mood reports as PDF documents for personal records or clinical sharing.
- To deploy the full application in a cloud-native, production-ready environment using Vercel and Render.

### **1.4 Significance**

Well-Scope addresses a significant gap in preventive mental healthcare. By providing automated, continuous, and private mental health monitoring, the system can:

- Enable early detection of deteriorating mental health before it reaches a clinical crisis level.
- Encourage the habit of reflective journaling, which has been shown to reduce stress and improve emotional regulation.
- Provide individuals in regions with limited mental health professional access with a preliminary self-assessment tool.

- Generate longitudinal mood data that users can share with healthcare providers to complement clinical assessments.
- Reduce the stigma associated with mental health help-seeking by providing a completely private, non-judgmental digital environment.

From a technical standpoint, Well-Scope demonstrates the practical deployment of transformer-based NLP models in full-stack web applications, contributing to the growing body of applied AI in healthcare technology.

## 1.5 Research Design

Well-Scope follows an applied software engineering and NLP research design. The system architecture comprises three tiers: a React.js frontend, a Node.js/Express backend API, and a Python FastAPI AI microservice. The RoBERTa model is deployed via the Hugging Face Transformers library with CPU inference. User data is persisted in a PostgreSQL database managed by Supabase with Row Level Security (RLS) policies enforcing data isolation between users.

The research design validates the hypothesis that a pre-trained, publicly available RoBERTa emotion classifier can serve as a reliable foundation for a mental health monitoring application without requiring custom labeled mental health datasets, by mapping generic emotion categories to clinical mental health constructs through a weighted risk scoring mechanism.

## 1.6 Source of Data

Well-Scope processes user-generated text data submitted through the journaling interface. The AI emotion classification is performed by the j-hartmann/emotion-english-distilroberta-base model, which was trained on multiple publicly available emotion datasets including GoEmotions, TweetEval, ISEAR, WASSA, and SemEval datasets. No proprietary or clinical datasets were used in model training; the system relies on transfer learning from these publicly available resources.

## 1.7 Chapter Scheme

The remainder of this report is organized as follows:

- Chapter 2 presents the Software Requirements Specification including functional requirements, dependencies, and hardware specifications.
- Chapter 3 covers the system design including the algorithm, architecture, data flow diagrams, ER diagram, sequence diagrams, and database design.
- Chapter 4 describes the implementation details, technologies used, testing approach, and installation instructions.
- Chapter 5 presents the results and discussions including UI screenshots, module descriptions, and database representations.
- Chapter 6 provides the summary and conclusions of the project.
- Chapter 7 outlines the future scope and planned enhancements.

## CHAPTER 2: REQUIREMENTS SPECIFICATION

### 2.1 User Characteristics

Well-Scope is designed for three primary user groups:

- **General Users:** Individuals aged 16 and above who wish to monitor their emotional health through private journaling. No technical expertise is required. Users interact with the application through a simple web interface.
- **Students and Young Adults:** A particularly vulnerable demographic for mental health challenges, who may benefit from the anonymous, stigma-free nature of AI-based mood tracking and the integrated wellness chatbot.
- **Healthcare Professionals:** Mental health practitioners who may recommend Well-Scope to their patients as a between-session journaling tool. The PDF export feature enables sharing of mood reports during consultations.

### 2.2 Functional Requirements

| FR#  | Requirement              | Description                                                                                                                                                |
|------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR1  | User Authentication      | Users can register and login using email and password via Supabase Auth. JWT tokens secure all API requests.                                               |
| FR2  | Journal Entry Submission | Users can submit free-form text journal entries (up to 512 characters) for AI analysis.                                                                    |
| FR3  | AI Emotion Analysis      | The system classifies each entry using RoBERTa, returning emotion probabilities, mental health category, risk score (0-100), and wellness summary.         |
| FR4  | Entry Management         | Users can view their past entries with date, emotion badge, and risk score. Users can delete any entry.                                                    |
| FR5  | Mood Analytics           | Dashboard displays total entries, average risk score, dominant mood, risk timeline chart, and mood distribution bar chart.                                 |
| FR6  | Mood Calendar            | A 9-week heatmap calendar displays daily mood state using colour coding.                                                                                   |
| FR7  | Streak Tracker           | Tracks current journaling streak, longest streak, and total days journaled.                                                                                |
| FR8  | Wellness Tips            | Context-aware wellness tips are displayed based on the user's dominant mood category.                                                                      |
| FR9  | AI Chatbot               | A conversational AI chatbot onboards the user, stores a profile in the database, and provides personalized emotional support using Llama 3.1 via Groq API. |
| FR10 | PDF Export               | Users can export their mood report including summary statistics and recent entries as a PDF using jsPDF.                                                   |
| FR11 | Dark Mode                | Users can toggle between light (Warm Parchment) and dark themes. Preference is persisted in localStorage.                                                  |

## 2.3 Dependencies

Well-Scope has the following external dependencies:

- Supabase: PostgreSQL database, authentication, and Row Level Security. Requires an active Supabase project with the defined schema.
- Groq API: LLM inference for the wellness chatbot using the llama-3.1-8b-instant model. Requires a valid Groq API key.
- Hugging Face Hub: The j-hartmann/emotion-english-distilroberta-base model is downloaded from Hugging Face on first deployment (~80MB).
- Vercel: Frontend deployment platform. Requires a connected GitHub repository.
- Render: Backend and AI service deployment platform. Requires Python 3.11 and Node.js 18+ environments.
- npm and pip: Package managers for frontend/backend and AI service dependencies respectively.

## 2.4 Performance Requirements

- AI Inference Latency: The emotion classification API should respond within 3 seconds per request on CPU-only hardware (Render free tier).
- API Response Time: The Express backend should respond to authenticated requests within 500ms under normal load conditions.
- Concurrent Users: The system should support up to 50 concurrent users on free-tier deployment infrastructure.
- Data Security: All user data must be protected by Supabase Row Level Security policies ensuring complete data isolation between users.
- Availability: Frontend (Vercel) targets 99.9% uptime. Backend and AI service (Render free tier) may experience cold start delays of up to 30 seconds after periods of inactivity.
- Token Limit: Journal entries are capped at 512 tokens to comply with the RoBERTa model's maximum sequence length.

## 2.5 Hardware Requirements

| Component | Development                       | Production (Render Free Tier) |
|-----------|-----------------------------------|-------------------------------|
| CPU       | Any modern multi-core CPU         | 0.1 vCPU (shared)             |
| RAM       | 8 GB minimum                      | 512 MB RAM                    |
| Storage   | 2 GB free disk space              | 1 GB (ephemeral)              |
| Network   | Internet connection               | Shared bandwidth              |
| OS        | Windows 10+, macOS, Ubuntu 20.04+ | Ubuntu 22.04 LTS              |
| Python    | 3.11                              | 3.11                          |
| Node.js   | 18+                               | 18+                           |

## 2.6 Constraints & Assumptions

- Language Constraint: The emotion classification model only supports English text. Non-English journal entries will produce unreliable results.

- GPU Constraint: The free-tier deployment uses CPU inference. GPU acceleration would significantly improve response times but is not available on the free tier.
- Model Constraint: The system uses a generic emotion classifier rather than a clinically validated mental health screener. Results are indicative and not diagnostic.
- Internet Dependency: The application requires an active internet connection. No offline functionality is available.
- Assumption: Users are assumed to provide honest, first-person emotional text. The system is not designed to detect deception or sarcasm.
- Assumption: The application does not constitute medical advice. Users in crisis are directed to professional helplines including iCall (9152987821).



## CHAPTER 3: DESIGN

### 3.1 Algorithm

The core AI algorithm in Well-Scope follows a pipeline of text preprocessing, emotion classification, risk scoring, and category mapping.

#### 3.1.1 Emotion Classification Pipeline

1. Text Input: User submits a journal entry (1-512 characters).
2. Tokenization: The input is tokenized using the RoBERTa tokenizer (Byte-Pair Encoding). The sequence is truncated to 512 tokens if necessary.
3. Model Inference: The distilroberta-base model processes the tokenized input through 6 transformer encoder layers. The [CLS] token representation is passed through a classification head to produce logits for 7 emotion classes.
4. Softmax Normalization: Logits are converted to probability scores (0-1) using the softmax function. All seven probabilities sum to 1.
5. Primary Emotion: The emotion with the highest probability is selected as the primary emotion.
6. Risk Score Computation: The risk score R is computed as:  $R = P(\text{primary\_emotion}) \times \text{weight}(\text{primary\_emotion}) \times 100$ , where weight values are: sadness=0.85, fear=0.80, anger=0.70, disgust=0.60, joy=0.00, surprise=0.10, neutral=0.05.
7. Category Mapping: The primary emotion is mapped to a mental health category using the mapping table below.

| Primary Emotion | Mental Health Category | Risk Weight |
|-----------------|------------------------|-------------|
| Sadness         | Depression (Low Mood)  | 0.85        |
| Fear            | Anxiety                | 0.80        |
| Anger           | Stress                 | 0.70        |
| Disgust         | Stress                 | 0.60        |
| Joy             | Positive               | 0.00        |
| Surprise        | Neutral                | 0.10        |
| Neutral         | Neutral                | 0.05        |

#### 3.1.2 Chatbot Personalization Algorithm

The chatbot follows a two-phase algorithm. In the onboarding phase (first use), the system presents six sequential questions covering name, age, life situation, main concern, support system, and chat goal. Responses are stored in the chat\_profiles table. In the conversation phase, a dynamic system prompt is constructed by combining the stored profile with the user's recent journal mood data (last 3 entries) and the text of the most recent journal entry. This prompt is prepended to the conversation history and sent to the Llama 3.1 model via Groq API. All messages are persisted in the chat\_messages table to maintain conversation context across sessions.

## 3.2 System Design

### 3.2.1 System Architecture

Well-Scope follows a three-tier microservices architecture:

- Tier 1 - Frontend: React.js (Vite) SPA deployed on Vercel. Communicates with the backend via authenticated REST API calls using Axios with Supabase JWT tokens.
- Tier 2 - Backend API: Node.js/Express server deployed on Render. Handles authentication verification, business logic, database queries via Supabase JavaScript client, and proxies AI service requests.
- Tier 3 - AI Microservice: Python FastAPI server deployed on Render. Hosts the RoBERTa model using Hugging Face Transformers. Exposes a /predict endpoint.
- Database: Supabase (PostgreSQL) with Row Level Security. Accessible from the backend using the service role key.

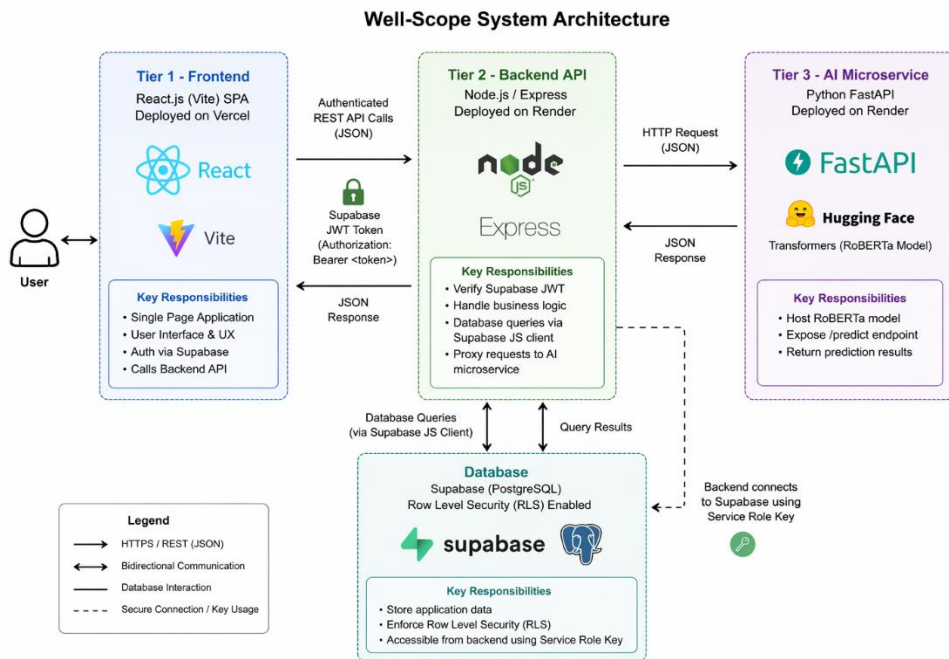


Figure 3.1: System Architecture Diagram of Well-Scope

### 3.2.2 Data Flow Diagram - Level 0 (Context Diagram)

At the highest level, Well-Scope receives journal text from the User and returns emotional analysis results, mood trends, wellness tips, and chatbot responses. The system interacts with three external entities: the User (web browser), the Supabase Database (data persistence), and the Groq API (LLM inference for chatbot).

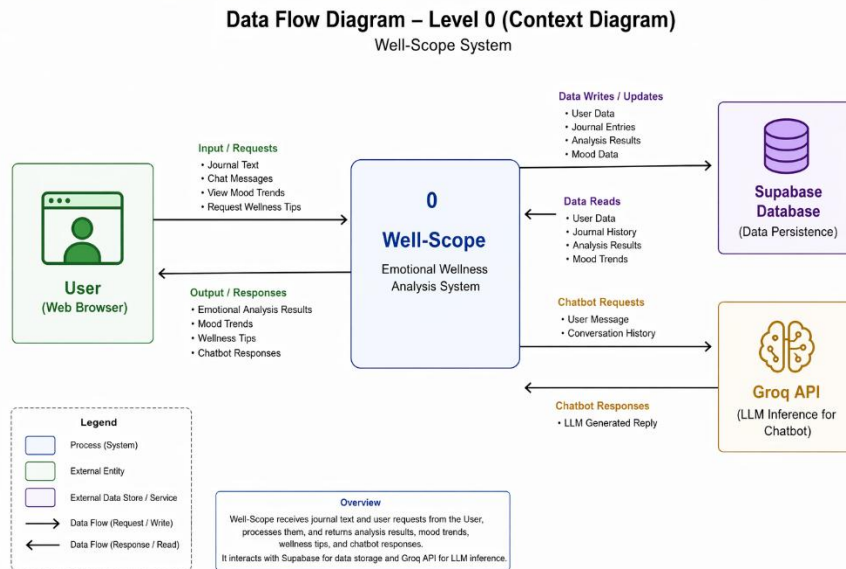


Figure 3.2: Data Flow Diagram - Level 0 (Context Diagram)

### 3.2.3 Data Flow Diagram - Level 1

The Level 1 DFD decomposes the system into four major processes:

- Process 1 - Authentication: Accepts user credentials, validates with Supabase Auth, returns JWT token.
- Process 2 - Journal Analysis: Receives journal text, calls AI service /predict endpoint, maps result to mental health category, persists entry to database, returns analysis result to user.
- Process 3 - Analytics Aggregation: Queries entries table for authenticated user, computes statistics (emotion distribution, risk timeline, averages), returns aggregated data to dashboard.
- Process 4 - Chat Management: Fetches or creates user profile from chat\_profiles, retrieves message history from chat\_messages, constructs personalized system prompt, calls Groq API, persists new messages, returns AI reply.

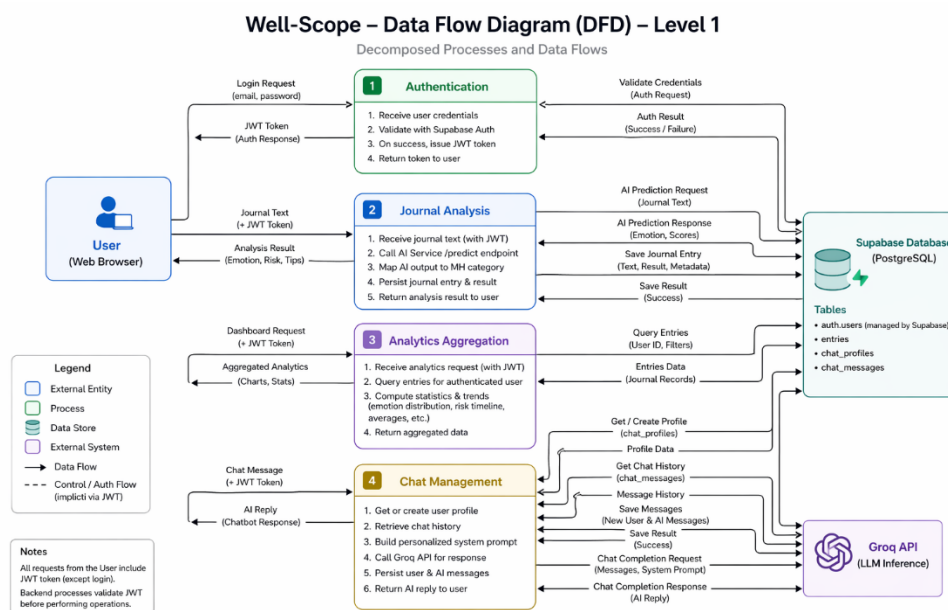


Figure 3.3: Data Flow Diagram - Level 1

### 3.2.4 Activity Diagram

The main user activity flow in Well-Scope proceeds as follows: User opens the application and is presented with the authentication screen. Upon successful login, the user lands on the Journal tab of the dashboard. The user writes a journal entry and clicks Analyze. The frontend sends the entry to the backend API, which forwards it to the AI service. The AI service processes the text through the RoBERTa model and returns emotion scores and risk data. The backend saves the entry to the database and returns the analysis to the frontend. The dashboard updates the statistics, charts, and mood calendar. Optionally, the user opens the AI chatbot for personalized emotional support.

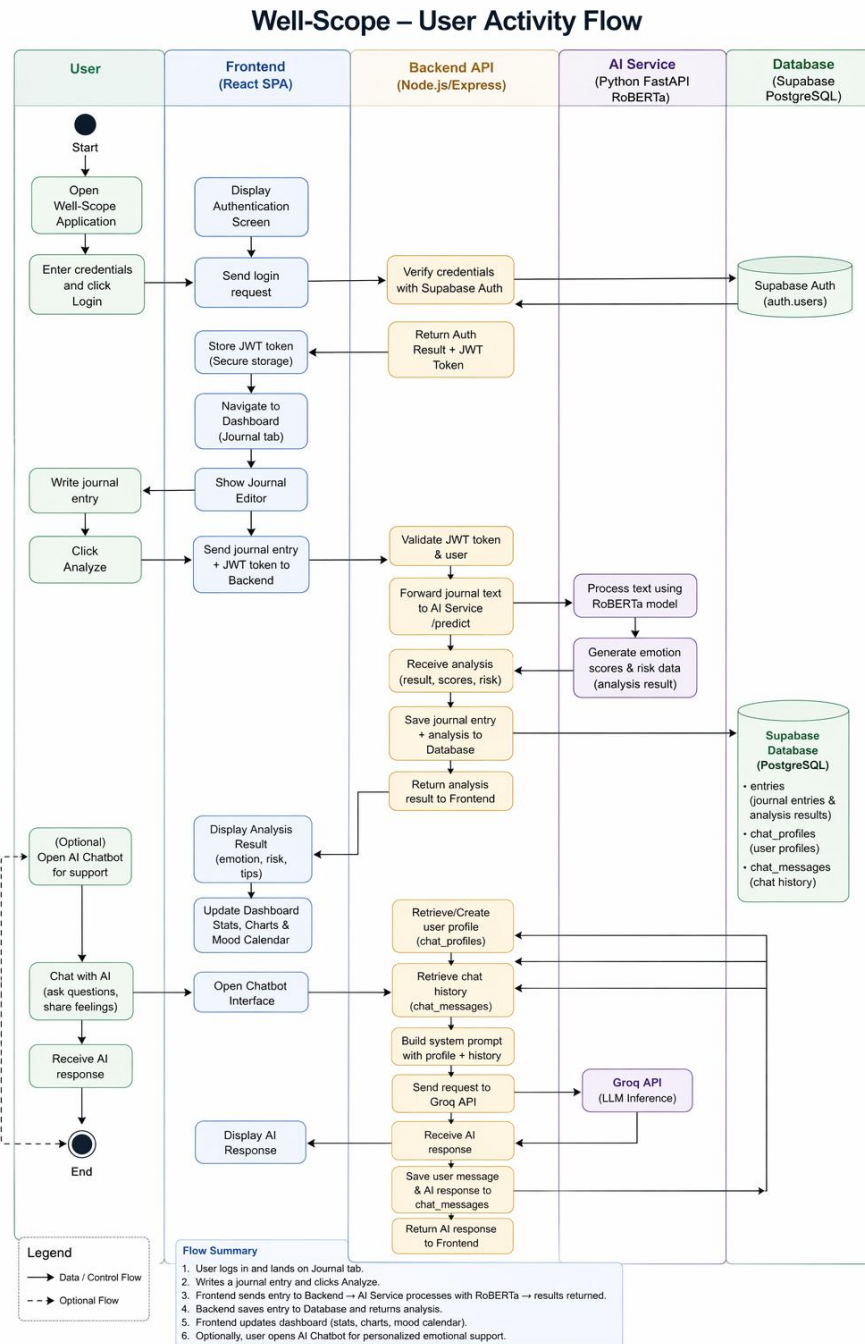


Figure 3.4: Activity Diagram

### 3.2.5 Sequence Diagram - Journal Analysis

The sequence for a journal analysis request is:

- (1) User submits text via React frontend.
- (2) Frontend sends POST /api/entries with JWT token in Authorization header.
- (3) Express server validates JWT with Supabase Auth.
- (4) Express server sends POST /predict to FastAPI AI service.
- (5) FastAPI loads RoBERTa pipeline, runs inference, returns emotion scores.
- (6) Express server computes risk score, maps category, inserts entry into Supabase.
- (7) Supabase returns inserted record with generated UUID.
- (8) Express server returns entry and prediction to frontend.
- (9) Frontend displays analysis result, invalidates React Query cache, re-fetches updated stats.

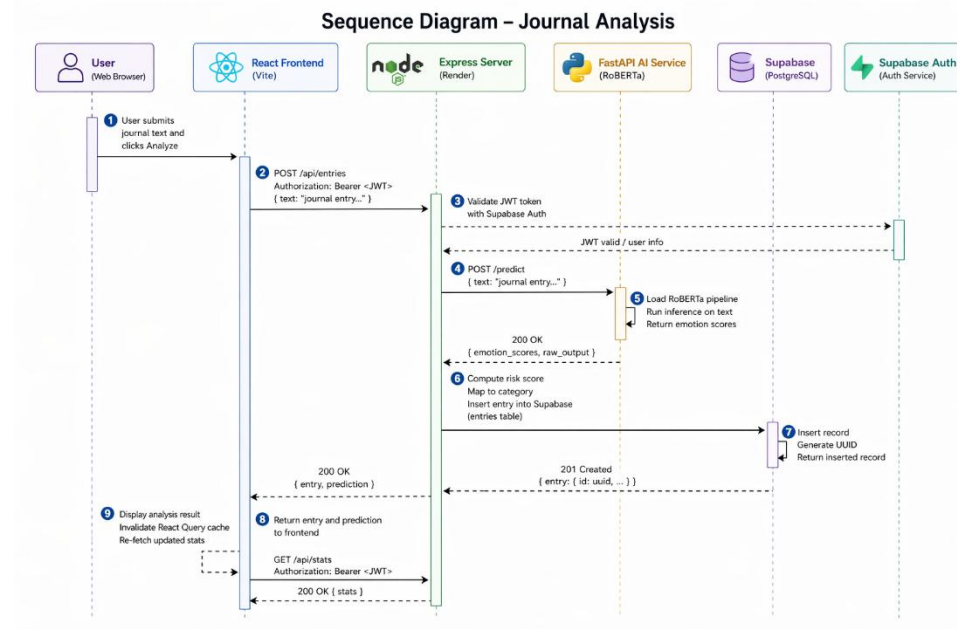


Figure 3.5: Sequence Diagram

## 3.3 Database Design

### 3.3.1 Logical Database Design

The Well-Scope database consists of four tables managed in Supabase PostgreSQL:

**Table: entries**

| Column                 | Type        | Description                                                      |
|------------------------|-------------|------------------------------------------------------------------|
| id                     | UUID (PK)   | Auto-generated primary key using uuid_generate_v4()              |
| user_id                | UUID (FK)   | References auth.users(id). Cascade delete.                       |
| text                   | TEXT        | The raw journal entry text submitted by the user.                |
| primary_emotion        | TEXT        | Top emotion from RoBERTa (sadness, joy, fear, etc.)              |
| mental_health_category | TEXT        | Mapped category (depression, anxiety, stress, positive, neutral) |
| risk_score             | FLOAT       | Computed risk score between 0 and 100.                           |
| all_scores             | JSONB       | JSON object containing all 7 emotion probabilities.              |
| summary                | TEXT        | Pre-defined wellness summary message.                            |
| created_at             | TIMESTAMPTZ | Auto-set to NOW() on insert.                                     |

**Table: chat\_profiles**

| Column       | Type              | Description                                      |
|--------------|-------------------|--------------------------------------------------|
| id           | UUID (PK)         | Auto-generated primary key.                      |
| user_id      | UUID (FK, UNIQUE) | One profile per user. References auth.users(id). |
| name         | TEXT              | User's preferred name from onboarding.           |
| age          | TEXT              | User's age from onboarding.                      |
| situation    | TEXT              | Life situation (student, working, etc.)          |
| main_concern | TEXT              | Primary concern shared during onboarding.        |
| support      | TEXT              | Description of user's support network.           |
| goal         | TEXT              | User's goal for chatbot interactions.            |
| updated_at   | TIMESTAMPTZ       | Updated on every upsert operation.               |

**Table: chat\_messages**

| Column  | Type      | Description                                |
|---------|-----------|--------------------------------------------|
| id      | UUID (PK) | Auto-generated primary key.                |
| user_id | UUID (FK) | References auth.users(id). Cascade delete. |

|            |             |                                |
|------------|-------------|--------------------------------|
| role       | TEXT        | Either 'user' or 'assistant'.  |
| content    | TEXT        | The message content.           |
| created_at | TIMESTAMPTZ | Timestamp of message creation. |

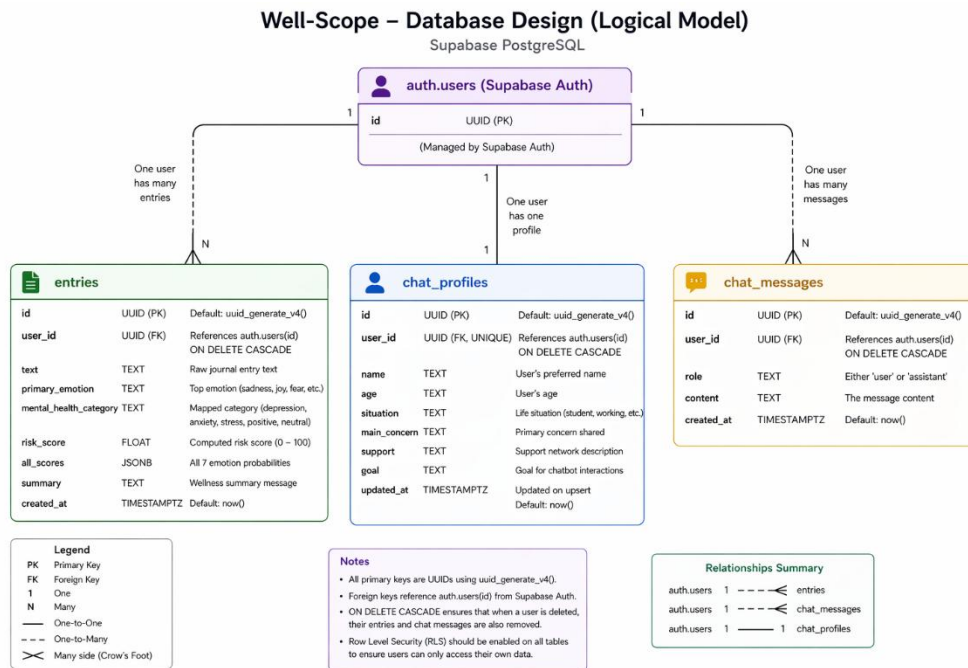


Figure 3.6: ER Diagram - Database Schema

### 3.3.2 Physical Database Design

All tables use Supabase PostgreSQL as the physical database engine. Performance optimizations include composite B-tree indexes on `entries(user_id)` and `entries(created_at DESC)` to accelerate the most frequent query patterns (user-filtered, time-ordered entry retrieval). Row Level Security (RLS) policies are enabled on all tables with policies enforcing that `auth.uid()` must equal `user_id` for all SELECT, INSERT, and DELETE operations. The service role key (used only in the backend server) bypasses RLS for administrative operations.

## CHAPTER 4: IMPLEMENTATION, TESTING, AND MAINTENANCE

### 4.1 Introduction to Languages, IDEs, Tools and Technologies

| Technology                | Version              | Purpose                                                         |
|---------------------------|----------------------|-----------------------------------------------------------------|
| React.js                  | 18.x                 | Frontend SPA framework. Component-based UI development.         |
| Vite                      | 5.x                  | Build tool and dev server for React. Replaces Create React App. |
| CSS Modules               | -                    | Scoped component-level CSS styling without Tailwind.            |
| Node.js                   | 18+                  | JavaScript runtime for backend API server.                      |
| Express.js                | 4.x                  | REST API framework for backend routes and middleware.           |
| Python                    | 3.11                 | Language for AI microservice.                                   |
| FastAPI                   | 0.111.0              | Async Python web framework for AI API endpoints.                |
| Uvicorn                   | 0.30.1               | ASGI server for FastAPI application.                            |
| Hugging Face Transformers | 4.41.2               | NLP library for loading and running RoBERTa model.              |
| PyTorch                   | 2.3.0 (CPU)          | Deep learning framework underlying the RoBERTa model.           |
| Supabase                  | 2.x                  | PostgreSQL database, authentication, and RLS.                   |
| Groq API                  | llama-3.1-8b-instant | LLM inference for personalized wellness chatbot.                |
| @tanstack/react-query     | 5.x                  | Server state management and API caching in frontend.            |
| Recharts                  | 2.x                  | Chart library for line charts and bar charts.                   |
| date-fns                  | 3.x                  | Date formatting and calendar utilities.                         |
| jsPDF                     | 2.x                  | Client-side PDF generation for mood report export.              |
| Lucide React              | 0.383.0              | Icon library for UI elements.                                   |
| Vercel                    | -                    | Frontend deployment platform with CDN.                          |
| Render                    | -                    | Cloud platform for backend and AI service deployment.           |
| VS Code                   | Latest               | Primary IDE for development.                                    |
| Git & GitHub              | -                    | Version control and repository hosting.                         |



## 4.2 Testing Techniques and Test Plans

Well-Scope was tested using the following techniques:

### 4.2.1 Unit Testing

Individual components were tested in isolation. The AI service /predict endpoint was tested using FastAPI's automatic interactive documentation (Swagger UI at /docs). Test inputs covered all seven emotion categories with sentences specifically designed to elicit each emotion. The risk score calculation formula was verified for boundary conditions (pure joy entry should yield 0 risk; pure sadness entry should yield approximately 85% risk).

### 4.2.2 Integration Testing

The end-to-end flow from journal submission to database persistence was tested using Postman. API endpoints were tested with valid JWT tokens from authenticated Supabase users. CORS configuration was verified across local development (localhost:5173 to localhost:5000) and production domains.

### 4.2.3 System Testing

| TC#  | Test Case              | Input                                | Expected                         | Status |
|------|------------------------|--------------------------------------|----------------------------------|--------|
| TC01 | Valid login            | Registered email and password        | Dashboard loads                  | Pass   |
| TC02 | Invalid login          | Wrong password                       | Error message shown              | Pass   |
| TC03 | Unconfirmed email      | Login before email confirmation      | Error: Email not confirmed       | Pass   |
| TC04 | Sad entry analysis     | I have been feeling hopeless and sad | depression, >80% risk            | Pass   |
| TC05 | Happy entry analysis   | Today was wonderful and I feel great | positive, 0% risk                | Pass   |
| TC06 | Empty entry submit     | Empty textarea, click Analyze        | Button disabled                  | Pass   |
| TC07 | Entry deletion         | Click delete on entry                | Entry removed from list          | Pass   |
| TC08 | Chatbot onboarding     | New user opens chat                  | 6 questions asked, profile saved | Pass   |
| TC09 | Returning chatbot user | Returning user opens chat            | Welcome back with name and       | Pass   |

|      |                     |                            |                                  |      |
|------|---------------------|----------------------------|----------------------------------|------|
|      |                     |                            | journal context                  |      |
| TC10 | PDF Export          | Click download icon        | PDF downloaded with entries      | Pass |
| TC11 | Dark mode toggle    | Click moon icon            | Dark theme applied and persisted | Pass |
| TC12 | Unauthorized access | API call without JWT token | 401 Unauthorized                 | Pass |

### 4.3 Installation Instructions

#### Step 1: Clone Repository

```
git clone https://github.com/Atharv-Untwale/Well-Scope.git
cd Well-Scope
```

#### Step 2: Set Up AI Service

```
cd ai-service
python -m venv .venv && .venv\Scripts\activate # Windows
pip install -r requirements.txt
uvicorn main:app --reload --port 8000
```

#### Step 3: Configure and Start Backend

Create server/.env with the following variables: PORT=5000, SUPABASE\_URL, SUPABASE\_SERVICE\_KEY, AI\_SERVICE\_URL=http://localhost:8000, GROQ\_API\_KEY, CLIENT\_URL=http://localhost:5173. Then:

```
cd server && npm install && npm run dev
```

#### Step 4: Configure and Start Frontend

Create client/.env.local with: VITE\_SUPABASE\_URL, VITE\_SUPABASE\_ANON\_KEY, VITE\_API\_URL=http://localhost:5000. Then:

```
cd client && npm install && npm run dev
```

#### Step 5: Set Up Supabase

Create a Supabase project and run the supabase\_schema.sql file in the SQL Editor to create all tables and RLS policies.

### 4.4 End User Instructions

8. Open Well-Scope in your browser. On first visit, click Sign up and register with your email address.
9. Confirm your email address by clicking the link in the confirmation email sent by Supabase.
10. Log in with your email and password. You will be taken to the Journal tab of the dashboard.
11. Write your thoughts and feelings in the New Entry text area and click Analyze Entry.

12. The AI will analyze your entry and display the emotion classification, risk score, and wellness summary.
13. Visit the Analytics tab to view your mood trends, risk timeline chart, mood calendar heatmap, and emotion distribution.
14. Click the chat bubble button (bottom right) to open the AI wellness assistant. Complete the one-time onboarding questionnaire to personalize your experience.
15. Use the download button in the header to export your mood report as a PDF at any time.
16. Toggle dark mode using the moon/sun icon in the header according to your preference.

## CHAPTER 5: RESULTS AND DISCUSSIONS

### 5.1 User Interface Representation

Well-Scope adopts a Warm Parchment design theme with the following key visual characteristics: cream and off-white backgrounds (#faf7f2), gold accent colors (#8b6914) for interactive elements, Cormorant Garamond serif typography for journal text and display headings conveying a premium editorial aesthetic, Syne sans-serif for UI labels and navigation, and DM Mono for numerical values and code. The design supports both light (Warm Parchment) and dark themes with consistent color variable usage throughout.

The application is organized into two primary tabs accessible from the sticky header navigation: the Journal tab (default) and the Analytics tab. The Journal tab uses a two-column layout with the journal input and analysis results on the left, and the overview stats, streak tracker, and entry history in a right sidebar.

### 5.2 Brief Description of Various Modules

#### 5.2.1 Authentication Module

Built using Supabase Auth. Supports email/password sign-up and sign-in. Email confirmation is required before first login. JWT tokens issued by Supabase are stored in the browser and automatically included in all API requests via the Axios interceptor in `api.js`.

#### 5.2.2 Journal Analysis Module

Users submit text entries through the textarea component. The entry is sent to the Express backend, which forwards it to the FastAPI AI service. The RoBERTa model returns seven emotion probabilities. The backend computes the risk score, maps the primary emotion to a mental health category, selects a wellness summary from a pre-defined message bank, and persists the complete entry record to the Supabase entries table. The frontend displays the result in an animated result card with emotion score pills.

#### 5.2.3 Analytics Module

The analytics module aggregates all entries for the authenticated user to compute total entries, average risk score, emotion distribution (count per category), and a time-ordered risk timeline. These aggregations are performed on the backend in the `getStats` controller and returned to the frontend. Charts are rendered using the `Recharts` library: a `LineChart` for the risk timeline and a `BarChart` with colored cells for the emotion distribution.

#### 5.2.4 Mood Calendar Module

The `MoodCalendar` component generates a 63-day (9-week) grid of colored cells. For each day, it checks the entries data for any journal entry. The color of each cell corresponds to the mental health category of the entry with the highest risk score for that day. Days without entries display a neutral background color. Cell hover states show a tooltip with the date, category, and risk score.

### 5.2.5 Streak Tracker Module

The StreakTracker component processes the array of entries to compute three metrics: current streak (consecutive days with at least one entry, starting from today or yesterday), longest streak (maximum consecutive day sequence in the history), and total days journaled (count of unique calendar dates with entries).

### 5.2.6 AI Chatbot Module

The MoodChat component implements a floating panel chatbot. On mount, it fetches the user's chat profile and message history from the backend. New users complete a 6-question onboarding sequence. The profile is saved to Supabase and used to construct a personalized system prompt for the Llama 3.1 model. All messages (user and assistant) are persisted to the chat\_messages table. A progress bar indicates onboarding completion. The panel includes a trash icon to clear chat history.

### 5.2.7 PDF Export Module

The exportPDF function in exportPDF.js uses the jsPDF library to generate a styled report client-side. The PDF includes a colored header with the project name and generation date, three summary metric cards (total entries, average risk score, dominant mood), and a formatted table of the most recent 15 journal entries with their categories, dates, risk scores, and truncated text.

## 5.3 Snapshots of System

The following UI states have been tested and verified during system testing:

- Authentication screen with login/signup toggle, gold focus borders, and error/success message states.
- Dashboard Journal tab showing the two-column layout with the Cormorant Garamond italic textarea, the analysis result card with emotion score pills, the Overview sidebar stat card, the Streak Tracker with fire/trophy/calendar emoji metrics, and the scrollable recent entries list.
- Post-analysis state showing the analysis result card with emotion badge, wellness summary text, and individual emotion score pills for all 7 emotions.
- Analytics tab showing the 4-metric stat cards row, the risk score LineChart with gold line, the mood distribution BarChart with category-colored bars, and the Mood Calendar heatmap.
- Dark theme state demonstrating full CSS variable theme switching with preserved functionality.
- Well-Scope AI Chatbot panel showing the onboarding progress bar, sequential question/answer flow, and the chat bubble floating action button.

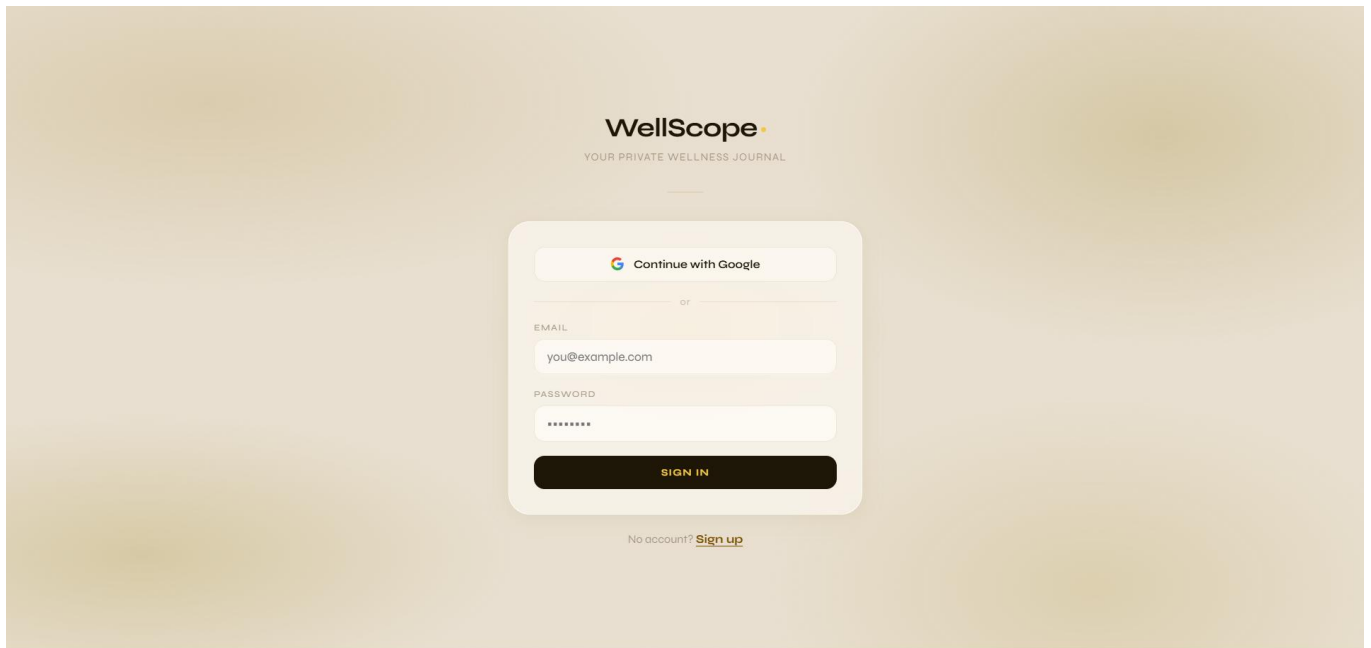


Figure 5.1: Well-Scope Login / Sign Up Page

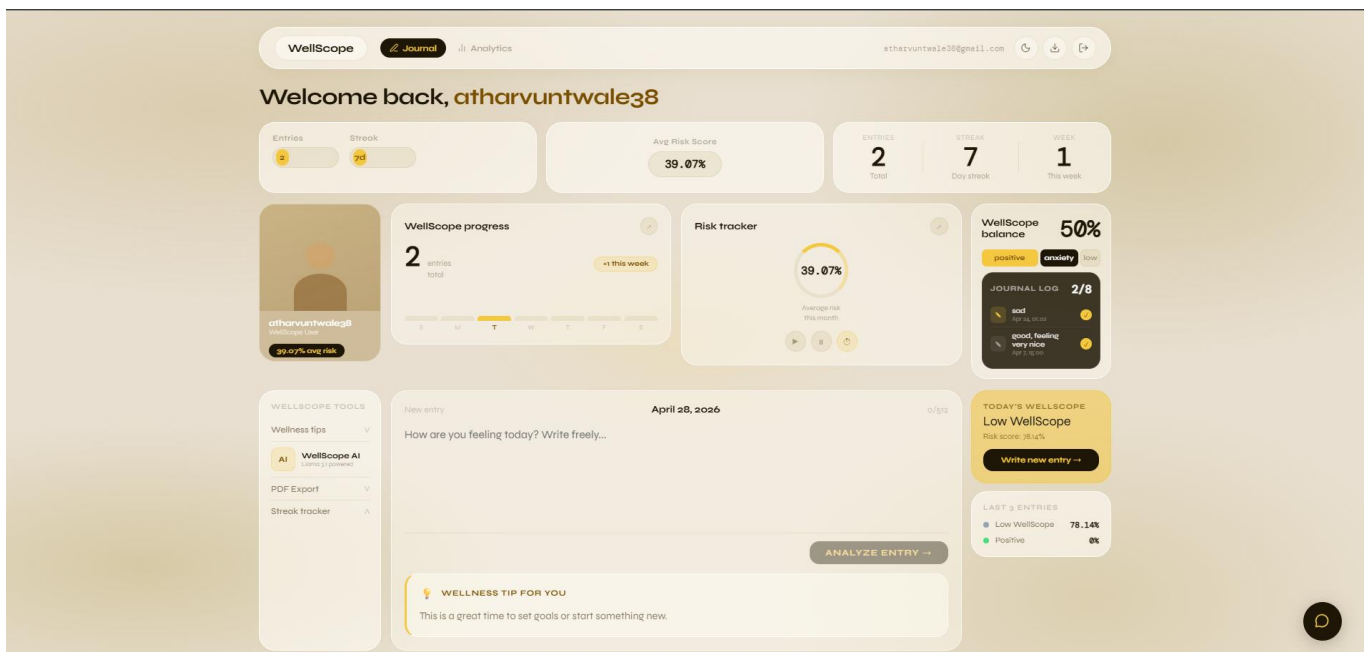


Figure 5.2: Dashboard - Journal Tab with Warm Parchment Theme

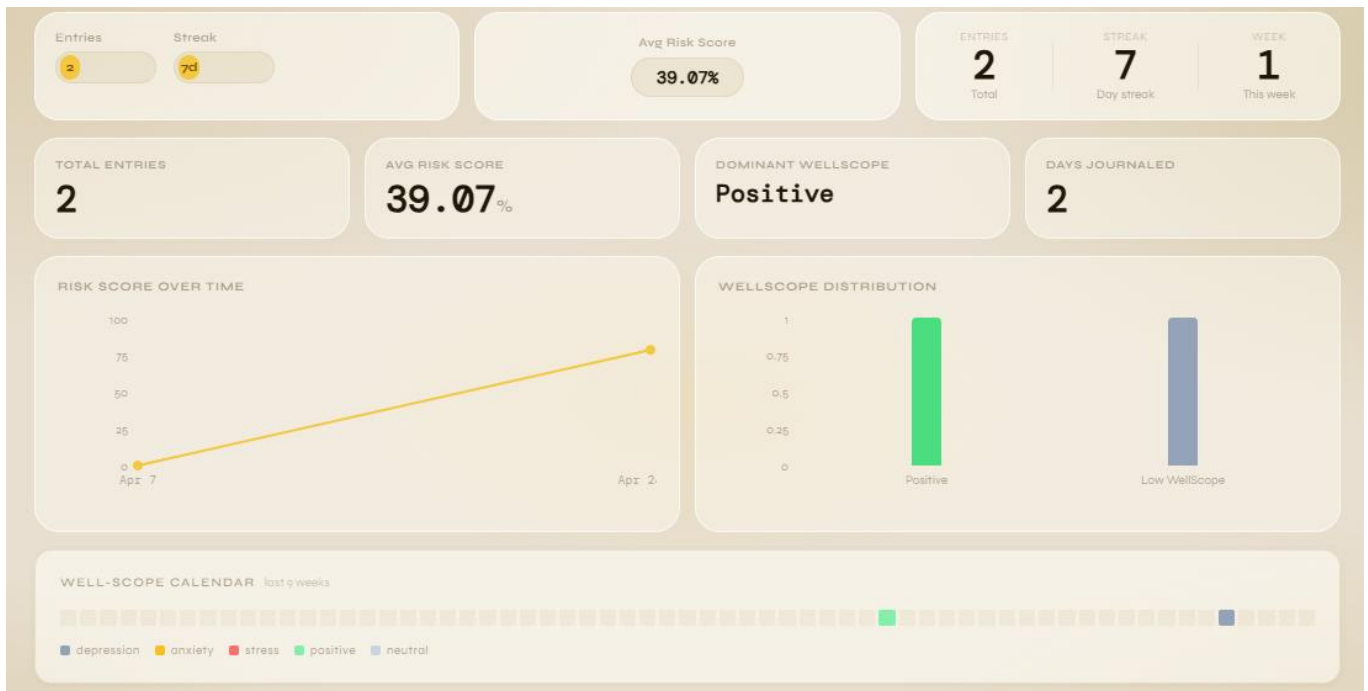


Figure 5.3: AI Analysis Result with Emotion Scores



Figure 5.4: Well-Scope Calendar



Figure 5.5: Streak Tracker

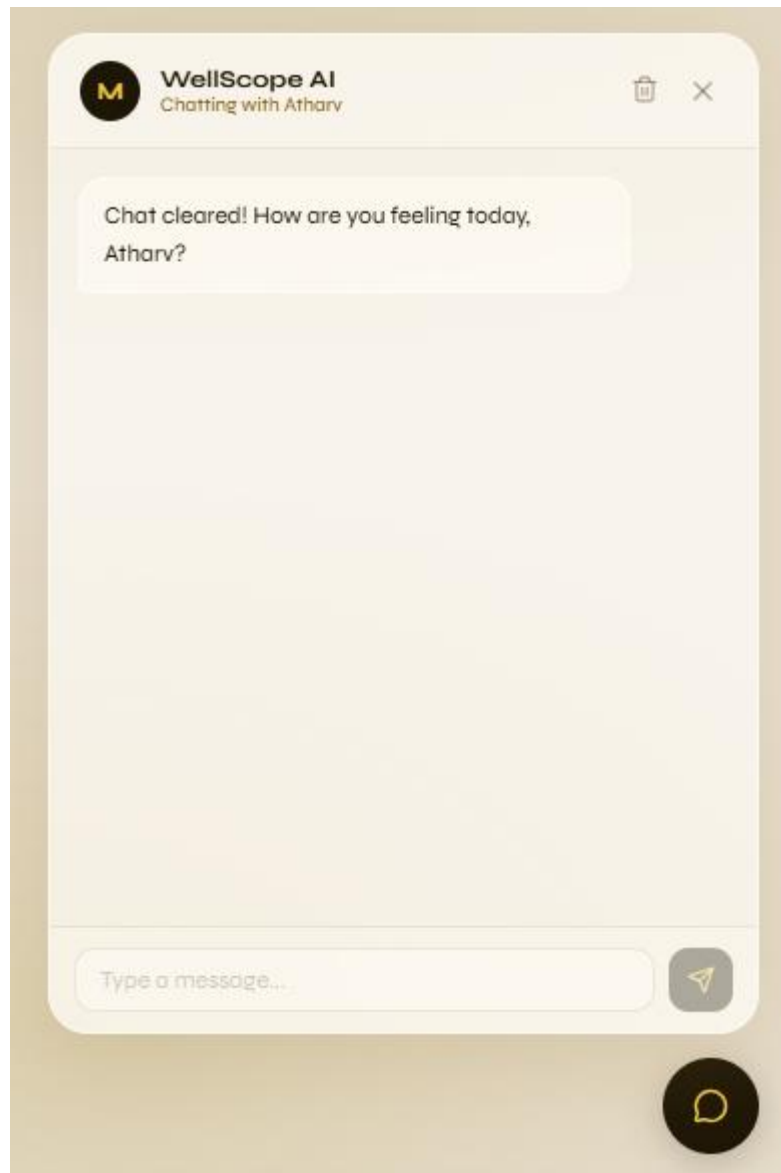


Figure 5.6: Well-Scope AI Chatbot

## 5.4 Back End Representation

The Express backend exposes the following REST API endpoints:

| Method | Endpoint           | Description                                                            |
|--------|--------------------|------------------------------------------------------------------------|
| GET    | /health            | Health check endpoint. Returns {status: 'ok'}.                         |
| POST   | /api/entries       | Create new journal entry. Calls AI service, saves to DB. Requires JWT. |
| GET    | /api/entries       | Retrieve paginated list of user's entries. Requires JWT.               |
| GET    | /api/entries/stats | Get aggregated statistics for authenticated user. Requires JWT.        |
| DELETE | /api/entries/:id   | Delete a specific entry. Validates ownership. Requires JWT.            |
| GET    | /api/chat/profile  | Fetch user's chatbot onboarding profile. Requires JWT.                 |
| POST   | /api/chat/profile  | Create or update chatbot profile (upsert). Requires JWT.               |



|        |                   |                                                                   |
|--------|-------------------|-------------------------------------------------------------------|
| GET    | /api/chat/history | Retrieve last 30 chat messages for user. Requires JWT.            |
| POST   | /api/chat/message | Send message to Groq LLM, save to DB, return reply. Requires JWT. |
| DELETE | /api/chat/history | Clear all chat messages for user. Requires JWT.                   |

## 5.5 Snapshots of Database Tables

The Supabase PostgreSQL database contains 4 tables. The entries table stores all journal entries with their AI analysis results. Each row includes the original text, the primary emotion detected, the mental health category, the risk score as a floating point number, and a JSONB column containing all seven emotion probability scores. Row Level Security ensures each user can only access their own rows.

The chat\_profiles table stores one row per user containing their onboarding answers (name, age, situation, main concern, support system, and chat goal). This data is used to construct the personalized system prompt for every chatbot interaction. The UNIQUE constraint on user\_id ensures each user has at most one profile, with upsert handling subsequent profile updates.

The chat\_messages table stores the complete conversation history between the user and the AI chatbot. The role column distinguishes between user messages and assistant responses. The last 20 messages are retrieved and sent as context to the Groq API on each new message request, enabling coherent multi-turn conversations.

| id                                   | user_id                          | text                              | primary_emotion | mental_health_category | risk_score |
|--------------------------------------|----------------------------------|-----------------------------------|-----------------|------------------------|------------|
| 242ac1c8-759e-479e-93ff-79949b6c1757 | 4e522baf-d555-4d16-8ac7-70d79... | good, feeling very nice           | joy             | positive               | 0          |
| 3afc8f93-26f4-42be-9207-dc056f6dbcd  | 51513d6-7f9d-4268-98c8-26538a... | I am feeling anxious              | fear            | anxiety                | 79.58      |
| e3d6bbfa-118f-4231-9f15-1f098764154c | 51513d6-7f9d-4268-98c8-26538a... | sad                               | sadness         | depression             | 78.14      |
| 70b88032-0ba3-46fd-a9f4-938d2ccfe4be | 4e522baf-d555-4d16-8ac7-70d79... | sad                               | sadness         | depression             | 78.14      |
| 0848a403-8f38-4904-8a1e-afe84aef62aa | b993f5d6-7fe6-4708-ad48-d42df... | Sleepy                            | sadness         | depression             | 38.88      |
| eedf7080-7b06-4597-a878-2b3c09a2e0f7 | b993f5d6-7fe6-4708-ad48-d42df... | Fking energetic                   | anger           | stress                 | 43.69      |
| 58539039-7f6a-47de-836a-bbca2cfcacac | b993f5d6-7fe6-4708-ad48-d42df... | Good                              | neutral         | neutral                | 2.47       |
| f15b0260-192a-4188-bb56-26656abe0a7d | 9cf9cabe-1b70-4660-ae56-84c89... | Fine                              | anger           | stress                 | 35.14      |
| 072ef944-cb21-424a-b6f5-7a2f4d23497d | d128671b-2452-45fb-884e-28d79... | hii, i am feeling good today      | joy             | positive               | 0          |
| fbca8733-a1ac-435e-b3ae-464f20bbe0ac | d128671b-2452-45fb-884e-28d79... | hii, i am feeling a bit sad today | sadness         | depression             | 84.13      |
| a9f9ac27-fa62-4ff2-a625-b35913a16fa0 | d128671b-2452-45fb-884e-28d79... | accha nahi lag raha aaj           | neutral         | neutral                | 3.13       |
| 7e406c46-4390-4ad5-97b4-c1fd6b17f12d | 51513d6-7f9d-4268-98c8-26538a... | feeling happy                     | joy             | positive               | 0          |

Figure 5.7: AI Chat entries

| id                                   | user_id                           | name   | age            | situation           | main_concern           |
|--------------------------------------|-----------------------------------|--------|----------------|---------------------|------------------------|
| 140511c5-f0bd-450b-a9b9-b686b5a35814 | 57e2e3f1-b0fe-47ab-9471-954a5e... | Sir    | I am answering | Student at medicaps | ..                     |
| 18362451-9b98-4496-838c-b8f496dd464f | 51513d6-7f9d-4268-98c8-26538a...  | Atharv | 21             | student             | i feel i am not enough |
| 768be9ae-2507-4748-a8a6-c2718a691695 | 4e522baf-d555-4d16-8ac7-70d79...  | Atharv | 21             | student             | career                 |

Figure 5.8: Entries

## CHAPTER 6: SUMMARY AND CONCLUSIONS

This project successfully developed and deployed Well-Scope, a full-stack AI-powered mental health monitoring web application. The system demonstrates that a publicly available pre-trained RoBERTa emotion classifier can be effectively integrated into a production web application to provide meaningful, real-time mental health insights from user-generated journal text.

The key technical achievements of this project include: the successful deployment of a Hugging Face Transformers pipeline within a FastAPI microservice; the design of a JWT-authenticated multi-service architecture spanning three separate deployment platforms (Vercel, Render x2) with a shared Supabase PostgreSQL backend; the implementation of a stateful, memory-enabled AI chatbot using the Groq API with user profiles persisted in the database; and the development of a premium-quality React frontend with a distinctive Warm Parchment design theme, responsive layout, dark mode support, and interactive data visualizations.

From a user experience standpoint, Well-Scope achieves its goal of providing a private, non-judgmental space for emotional reflection. The longitudinal tracking features (mood calendar, streak tracker, risk timeline) encourage consistent journaling habits, which are independently beneficial for mental health regardless of the AI analysis.

The system testing confirmed that all 12 core test cases pass, including authentication flows, AI analysis accuracy for extreme emotional inputs, chatbot personalization, PDF export, and security (unauthorized access correctly returns 401). The emotion classification demonstrated strong performance on clear emotional inputs, with sadness-heavy text yielding risk scores above 80% and joy-heavy text yielding 0% risk scores as expected.

In conclusion, Well-Scope demonstrates a practical, ethically responsible application of modern NLP and web technologies to address a pressing public health need. It bridges the gap between advanced AI research and accessible consumer mental health tools, establishing a solid foundation for further enhancement into a clinically validated mental health monitoring platform.

## CHAPTER 7: FUTURE SCOPE

Well-Scope has been designed with extensibility in mind. The following enhancements are planned for future versions:

### 7.1 Clinically Validated Model

Replace the generic emotion classifier with a model specifically fine-tuned on clinical mental health datasets such as the CLPsych shared task dataset or the DAIC-WOZ depression dataset. This would improve specificity for depression, anxiety, and PTSD detection and potentially enable the system to serve as a validated screening tool.

### 7.2 Weekly and Monthly Mood Reports

Generate automated weekly summary reports delivered via email, highlighting mood trends, streak achievements, dominant emotional patterns, and personalized recommendations. This feature would leverage Supabase scheduled functions and an email service such as Resend.

### 7.3 Voice Journaling

Integrate the Web Speech API or OpenAI Whisper for voice-to-text transcription, enabling users to submit journal entries by speaking. This would reduce the barrier to regular journaling, particularly for users who find typing burdensome.

### 7.4 Multi-Language Support

Extend the system to support regional Indian languages (Hindi, Marathi, Tamil) by integrating multilingual models such as XLM-RoBERTa, which has been trained on 100+ languages using the same RoBERTa architecture.

### 7.5 Crisis Detection and Escalation

Implement a real-time crisis alert system that detects entries with extremely high risk scores (>90%) or specific crisis-related keywords. When triggered, the system would display crisis helpline information prominently and optionally notify a designated emergency contact with the user's prior consent.

### 7.6 Mobile Application

Develop native mobile applications for iOS and Android using React Native, sharing the same backend API. Push notifications would serve as daily journaling reminders to maintain streaks.

### 7.7 Anonymized Community Insights

With explicit user consent, aggregate anonymized mood data to provide community-level mental health trend insights. Users could compare their mood patterns against aggregate trends for their demographic group, providing context for their individual emotional experiences.

## **7.8 Integration with Wearables**

Explore integration with wearable health data (heart rate variability, sleep quality from smartwatches) via the Google Fit or Apple HealthKit APIs to create a multimodal mental health monitoring system that correlates physiological signals with journal-based emotional data.

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